

Hassan Mostafa

associate professor

基本信息

姓名: Hassan Mostafa

所属院校: 开罗大学

地区: Egypt

年龄范围: 40岁至50岁

海外经历: 有

学科领域

通信工程

集成电路设计与集成系统

电子与计算机工程

微电子科学与工程

研究领域

FPGA与可重构计算

通信系统仿真与优化

低功耗电路与能量管理

无线通信与智能天线

低功耗设计与能耗优化

光纤通信与光网络

EDA工具与电路验证

移动通信与5G/6G

微电子器件与制造

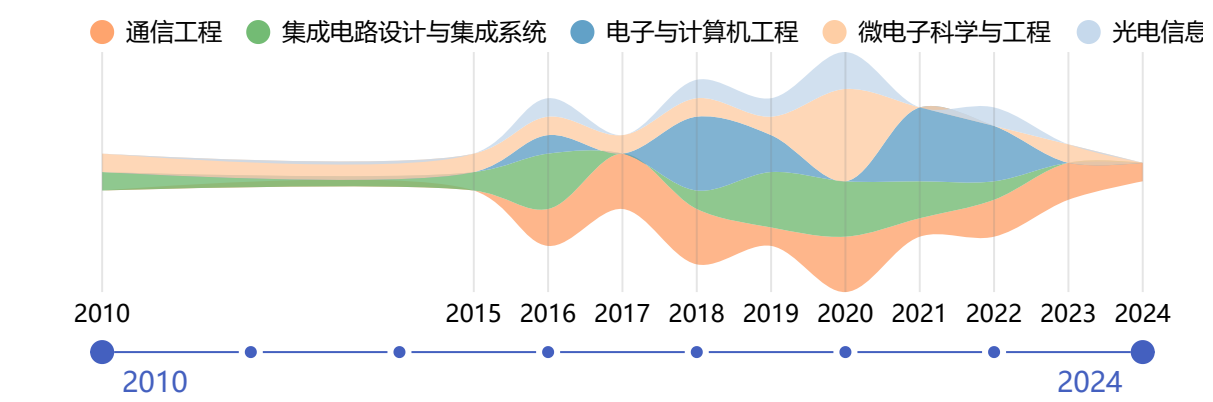
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个人简介

Dr. Hassan Mostafa (S' 01, M'11, SM'15) is an Assistant Professor of Nanoelectronics, Bioelectronics, and Optoelectronics at Cairo University and Zewail City of Science and Technology. He obtained his B.Sc., M.Sc. from Cairo University in 2001 and 2005, and a PhD from the University of Waterloo in 2011. He has worked as a post - doctoral fellow at the University of Toronto and with IMEC. Collaborating with many global sponsors, he has over 140 papers and 5 books. His research interests are diverse. He's a member of IEEE TC - VSA since 2017 and an IEEE Senior Member since 2015, and has won multiple awards.

近年发表的论文



Invited Paper: A Pseudo-Differential Architecture for Low-Power Voltage-to-Time Converters
2024 IEEE 67TH INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS, MWSCAS 2024
Hassan, Ali H.; Mostafa, Hassan; Yang, Chih-Kong Ken
被引用次数: 0 2024

A Case Study in Applying Functional Safety to an A* Accelerator Design
4TH INTERDISCIPLINARY CONFERENCE ON ELECTRICS AND COMPUTER, INTCEC 2024
Osama, Adham; Mostafa, Ahmed; Gamal, Mohamed; Imam, Usama; Khalil, Ahmed; Mostafa, Hassan; Mamdouh, Eslam; Ahmed, Islam; Taha, Mohamed
被引用次数: 0 2024

Beamwidth Design Tradeoffs in Radar-Aided Millimeter-Wave Cellular Networks: A Stochastic Geometry Approach
IEEE ACCESS
Nabil, Yasser; ElSawy, Hesham; Al-Dharrab, Suhail; Mostafa, Hassan; Attia, Hussein
被引用次数: 0 2024

Ultra-Reliable Device-Centric Uplink Communications in Airborne Networks: A Spatiotemporal Analysis
IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY
Nabil, Yasser; ElSawy, Hesham; Al-Dharrab, Suhail; Attia, Hussein; Mostafa, Hassan
被引用次数: 2 2023

A Stochastic Geometry Analysis for Joint Radar Communication System in Millimeter-wave Band
ICC 2023-IEEE INTERNATIONAL CONFERENCE ON COMMUNICATIONS
Nabil, Yasser; ElSawy, Hesham; Al-Dharrab, Suhail; Attia, Hussein; Mostafa, Hassan
被引用次数: 2 2023

Low Power Microarchitecture Designs of ACS Block in Viterbi Decoder: A Review
13TH INTERNATIONAL CONFERENCE ON INFORMATION COMMUNICATION AND MANAGEMENT, ICICM 2023

Mosbeh, Asmaa; Ibraheem, Ali A. Y.; Mostafa, Hassan; Yousef, Khalil

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2023

UAV-Aided Data Acquisition Using Gaining-Sharing Knowledge Optimization Algorithm

CMC-COMPUTERS MATERIALS & CONTINUA

Tawfik, Rania M.; Nomer, Hazem A. A.; Darweesh, M. Saeed; Mohamed, Ali Wagdy; Mostafa, Hassan

被引用次数: 3

2022

Implementation of a DPU-Based Intelligent Thermal Imaging Hardware Accelerator on FPGA

ELECTRONICS

Hussein, Abdelrahman S.; Anwar, Ahmed; Fahmy, Yasmine; Mostafa, Hassan; Salama, Khaled Nabil; Kafafy, Mai

被引用次数: 6

2022

A Novel Refreshment Circuit for 2T1M Neuromorphic Synapse

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Goda, Mai M.; Hassan, Ahmed H.; Mostafa, Hassan; Soliman, Ahmed M.

被引用次数: 2

2022

Radio Optical Network Simulation Tool (RONST)

CMC-COMPUTERS MATERIALS & CONTINUA

Abdelhak, Yasmine, I; Kamel, Fady; Hafez, Moustafa; Kotb, Hussein E.; Omran, Haitham A.; Ismail, Tawfik; Mostafa, Hassan

被引用次数: 0

2022

Energy-Efficient Precision-Scaled CNN Implementation With Dynamic Partial Reconfiguration

IEEE ACCESS

Youssef, Eman; Elsimary, Hamed A.; El-Moursy, Magdy A.; Mostafa, Hassan; Khattab, Ahmed

被引用次数: 5

2022

Towards a Generic UVM

2022 IEEE HIGH PERFORMANCE EXTREME COMPUTING VIRTUAL CONFERENCE (HPEC)

Mahmoud, Kholoud; Ahmed, Randa; Ayman, Karim; Ayman, Mostafa; Taie, Waleed; Ibrahim, Yasser; Mostafa, Hassan; Salah, Khaled

被引用次数: 1

2022

Generation of low-complexity power-efficient IR-UWB waveforms

OPTIK

Soliman, Samy S.; Shehata, Mohamed; Mostafa, Hassan

被引用次数: 4

2022

Power Efficient Design of High-Performance Convolutional Neural Networks Hardware Accelerator on FPGA: A Case Study With GoogLeNet

IEEE ACCESS

Abd El-Maksoud, Ahmed J.; Ebbad, Mohamed; Khalil, Ahmed H.; Mostafa, Hassan

被引用次数: 5

2021

A Novel Companding Technique to Reduce High Peak to Average Power Ratio in OFDM Systems

IEEE ACCESS

Mohammed, Abdulwahid; Ismail, Tawfik; Nassar, Amin; Mostafa, Hassan

被引用次数: 10

2021

Study of the Impact of Design Factors on Micro-Fluxgate Sensing Element

2021 IEEE INTERNATIONAL CONFERENCE ON SENSORS AND NANOTECHNOLOGY (SENNANO)

Elkattan, Mohamed; Mostafa, Hassan; Khalil, A. H.

被引用次数: 0

2021

A 0.002-mm² 8-bit 1-MS/s low-power time-based DAC (T-DAC)

IET CIRCUITS DEVICES & SYSTEMS

Hassan, Ali H.; Mostafa, Hassan; Refky, Mohamed; Salama, Khaled N.; Soliman, Ahmed M.

被引用次数: 2

2021

Functional Verification of Dynamic Partial Reconfiguration for Software-Defined Radio

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Ahmed, Islam; Mohieldin, Ahmed Nader; Mostafa, Hassan

被引用次数: 0

2021

Design of a reconfigurable network-on-chip for next generation FPGAs using Dynamic Partial Reconfiguration

MICROELECTRONICS JOURNAL

Ramy, Ahmed; Mostafa, Hassan; Khalil, A. H.

被引用次数: 0

2021

Responsivity of Fluxgate Sensing Element Under Different Excitation Conditions

2020 8TH INTERNATIONAL CONFERENCE ON INTELLIGENT AND ADVANCED SYSTEMS (ICIAS)

Elkattan, Mohamed; Mostafa, Hassan; Khalil, A. H.

被引用次数: 0

2021

Implementation of deep neural networks on FPGA-CPU platform using Xilinx SDSOC

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Hassan, Rania O.; Mostafa, Hassan

被引用次数: 4

2021

Accurate Closed Form Expressions for The Time-Bandwidth Product and FCC Violation Ratio as New Metrics To Characterize UWB Optical Systems

OPTIK

Aly, Eslam A.; Alabd, Mostafa B.; Kassem, Nouran; Mustafa, Waleed; Shehata, Mohamed; Mostafa, Hassan

被引用次数: 1

2020

Design of the Baseband Physical Layer of NarrowBand IoT LTE Uplink Digital Transmitter

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Mohamed, Basma H.; Taha, Ahmed; Shawky, Ahmed; Ahmed, Essraa; Mohamed, Ali; Mohsen, Manar; Samy, Rodina; ELHosiny, Amr; Ibrahim, Ahmed; Mostafa, Hassan

被引用次数: 2

2020

EANN: Energy Adaptive Neural Networks

ELECTRONICS

Hassan, Salma; Attia, Sameh; Salama, Khaled Nabil; Mostafa, Hassan

被引用次数: 5

2020

A Novel Highly Linear Voltage-To-Time Converter (VTC) Circuit for Time-Based Analog-To-Digital Converters (ADC) Using Body Biasing

ELECTRONICS

Elgreatly, Ahmed; Dessouki, Ahmed; Mostafa, Hassan; Abdalla, Rania; El-Rabaie, El-sayed

被引用次数: 1

2020

A high-efficiency piezoelectric-based integrated power supply for low-power platforms

MICROELECTRONICS JOURNAL

Hassan, Alaa; El-Shaboury, Aya; Mohamed, Karim; Askar, Mohamed; Mowafy, Osama; Mosaad, Yasmine; Hassan, Ali H.; Mostafa, Hassan

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2020

Maximum emission levels of photonicallly generated impulse radio waveforms under spectral constraints

OPTIK

Shehata, Mohamed; Abotaleb, Abdelrhman M.; Ali, Moaz G.; Hameed, Mohamed Farhat O.; Mostafa, Hassan; Obayya, Salah S. A.

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2020

A theoretical modeling analysis for triboelectrification controlled light emitting diodes

NANO ENERGY

Ahmed, Abdelsalam; Shehata, Mohamed; Hassan, Islam; Abdelhak, Yasmine, I; Cigdem, Efe; El-Kady, Maher; Ismail, Yehea; Mostafa, Hassan

被引用次数: 4

2020

Automated performance-based design technique for an efficient LTE PDSCH implementation using SDSoC tool

INTERNATIONAL JOURNAL OF COMMUNICATION SYSTEMS

Eladawy, Mohamed; Mostafa, Mahmoud; Sameh Said, M.; Mostafa, Hassan

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2020

A high precision write/read circuits for memristors using digital input/output interfaces

MICROELECTRONICS JOURNAL

Daoud, Ahmad A.; Dessouki, Ahmed S.; Mostafa, Hassan; Abdallah, Rania M.; El-Rabaie, El-Sayed M.

被引用次数: 10

2020

An accurate model of domain-wall-based spintronic memristor

INTEGRATION-THE VLSI JOURNAL

Nafea, Sherif F.; Dessouki, Ahmed A. S.; El-Rabaie, S.; Elnaghi, Basem E.; Ismail, Yehea; Mostafa, Hassan

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2019

A Generalized Framework for the Performance Evaluation of Microwave Photonic Assisted IR-UWB Waveform Generators

IEEE SYSTEMS JOURNAL

Shehata, Mohamed; Said, Mohamed Sameh; Mostafa, Hassan

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2019

Power adaptive high-resolution neural data compression algorithm (PANDCA)

MICROELECTRONICS JOURNAL

Ashraf, Mohammed; Mostafa, Hassan; Eladawy, Ahmed; Ismail, Yehea

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2019

Design exploration for network on chip based FPGAs: 2D and 3D tiles to router interface

MICROELECTRONICS JOURNAL

Salaheldin, Alaa; Mostafa, Hassan; Soliman, Ahmed M.

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2019

FPGA implementation of dynamically reconfigurable IoT security module using algorithm hopping

INTEGRATION-THE VLSI JOURNAL

Soliman, Shady; Jaala, Mohammed A.; Abotaleb, Abdelrhman M.; Hassan, Youssef; Abdelghany, Mohamed A.; Abdel-Hamid, Amr T.; Salama, Khaled N.; Mostafa, Hassan

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2019

ASIC and FPGA Comparative Study for IoT Lightweight Hardware Security Algorithms

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Samir, Nagham; Hussein, Abdelrahman Sobeih; Khaled, Mohaned; El-Zeiny, Ahmed N.; Osama, Mahetab; Yassin, Heba; Abdelbaky, Ali; Mahmoud, Omar; Shawky, Ahmed; Mostafa, Hassan

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2019

Optimizing FPGA-based hard networks-on-chip by minimizing and sharing resources

INTEGRATION-THE VLSI JOURNAL

Attia, Sameh; Fahmy, Hossam A. H.; Ismail, Yehea; Mostafa, Hassan

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2018

Dual Notched Band Quad-Element MIMO Antenna With Multitone Interference Suppression for IR-UWB Wireless Applications

IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION

Shehata, Mohamed; Said, Mohamed Sameh; Mostafa, Hassan

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2018

Comparative mechanical analysis of deep brain stimulation electrodes

BIOMEDICAL ENGINEERING ONLINE

Draz, H. H.; Gabran, S. R. I.; Basha, Mohamed; Mostafa, Hassan; Abu-Elyazeed, Mohamed F.; Zaki, Amal

Technology Scaling Roadmap for FinFET-Based FPGA Clusters Under Process Variations

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Abdelkader, Osama; El-Din, Mohamed Mohie; Mostafa, Hassan; Abdelhamid, Hamdy; Fahmy, Hossam A. H.; Ismail, Yehea; Soliman, Ahmed M.

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2018

On the Theoretical Limits of the Power Efficiency of Photonically Generated IR-UWB Waveforms

JOURNAL OF LIGHTWAVE TECHNOLOGY

Shehata, Mohamed; Mostafa, Hassan; Ismail, Yehea

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2018

Dynamic power estimation using Transaction Level Modeling

MICROELECTRONICS JOURNAL

Baher, Amr; El-Zeiny, Ahmed N.; Aly, Ahmed; Khalil, Ahmed; Hassan, Adham; Saeed, AbdelRahman; El Makarem, Karim Abo; El-Moursy, Magdy; Mostafa, Hassan

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2018

A new design methodology for voltage-to-frequency converters (VFCs) circuits suitable for time-based analog-to-digital converters (T-ADCs)

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

ElGabry, Mohammed A.; Hassan, Ali H.; Mostafa, Hassan; Soliman, Ahmed M.

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2018

Dual Split-Merge: A high throughput router architecture for FPGAs

MICROELECTRONICS JOURNAL

Helal, Khaled; Attia, Sameh; Fahmy, Hossam A. H.; Ismail, Tawfik; Ismail, Yehea; Mostafa, Hassan

被引用次数: 5

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NoC-DPR: A new simulation tool exploiting the Dynamic Partial Reconfiguration (DPR) on Network-on-Chip (NoC) based FPGA

INTEGRATION-THE VLSI JOURNAL

Hassan, Amr; Mostafa, Hassan; Fahmy, Hossam A. H.

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2018

A Novel MIM-Capacitor-Based 1-GS/s 14-bit Variation-Tolerant Fully-Differential Voltage-to-Time Converter (VTC) Circuit

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

El-Bayoumi, Abdullah; Mostafa, Hassan; Soliman, Ahmed M.

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2017

Towards the implementation of Multi-band Multi-standard Software-Defined Radio using Dynamic Partial Reconfiguration

INTERNATIONAL JOURNAL OF COMMUNICATION SYSTEMS

Sadek, Ahmad; Mostafa, Hassan; Nassar, Amin; Ismail, Yehea

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2017

Accurate Closed-Form Expressions for the Bit Rate-Wireless Transmission Distance Relationship in IR-UWBoF Systems

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Shehata, Mohamed; Mostafa, Hassan; Ismail, Yehea

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2017

A Novel 10-Bit 2.8-mW TDC Design Using SAR With Continuous Disassembly Algorithm

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ragab, Karim O.; Mostafa, Hassan; Eladawy, Ahmed

被引用次数: 18

2016

Electrical and optical clock and data recovery in optical access networks: a comparative study

INTERNATIONAL JOURNAL OF COMMUNICATION SYSTEMS

Abd El-Khaleq, Esraa; El-Sayed, Yasmine; Ismail, Tawfik; Mostafa, Hassan

被引用次数: 1

2016

Process Variation Aware Design of Multi-Valued Spintronic Memristor-Based Memory Arrays

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Mostafa, Hassan; Ismail, Yehea

被引用次数: 18

2016

On the use of a programmable front-end for multi-band/multi-standard applications

MICROELECTRONICS JOURNAL

Abdelsalam, Hoda; Hegazi, Emad; Mostafa, Hassan; Ismail, Yehea

被引用次数: 0

2016

Statistical yield improvement under process variations of multi-valued memristor-based memories

MICROELECTRONICS JOURNAL

Mostafa, Hassan; Ismail, Yehea

被引用次数: 13

2016

A design-oriented timing jitter/skew model in voltage-to-time converter (VTC) circuits

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Mostafa, Hassan; Ismail, Yehea I.

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2015

Micro-scale variation-tolerant exponential tracking energy harvesting system for wireless sensor networks

MICROELECTRONICS JOURNAL

Eltaliawy, Ayman; Mostafa, Hassan; Ismail, Yehea

被引用次数: 6

2015

A new current mode implementation of a balanced-output-signal generator

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Ghallab, Yehya H.; Mostafa, Hassan; Ismail, Yehea

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2014

NOVEL FCS-BASED LAYOUT-FRIENDLY ACCURATE WIDE-BAND LOW-POWER CCII-
REALIZATIONS

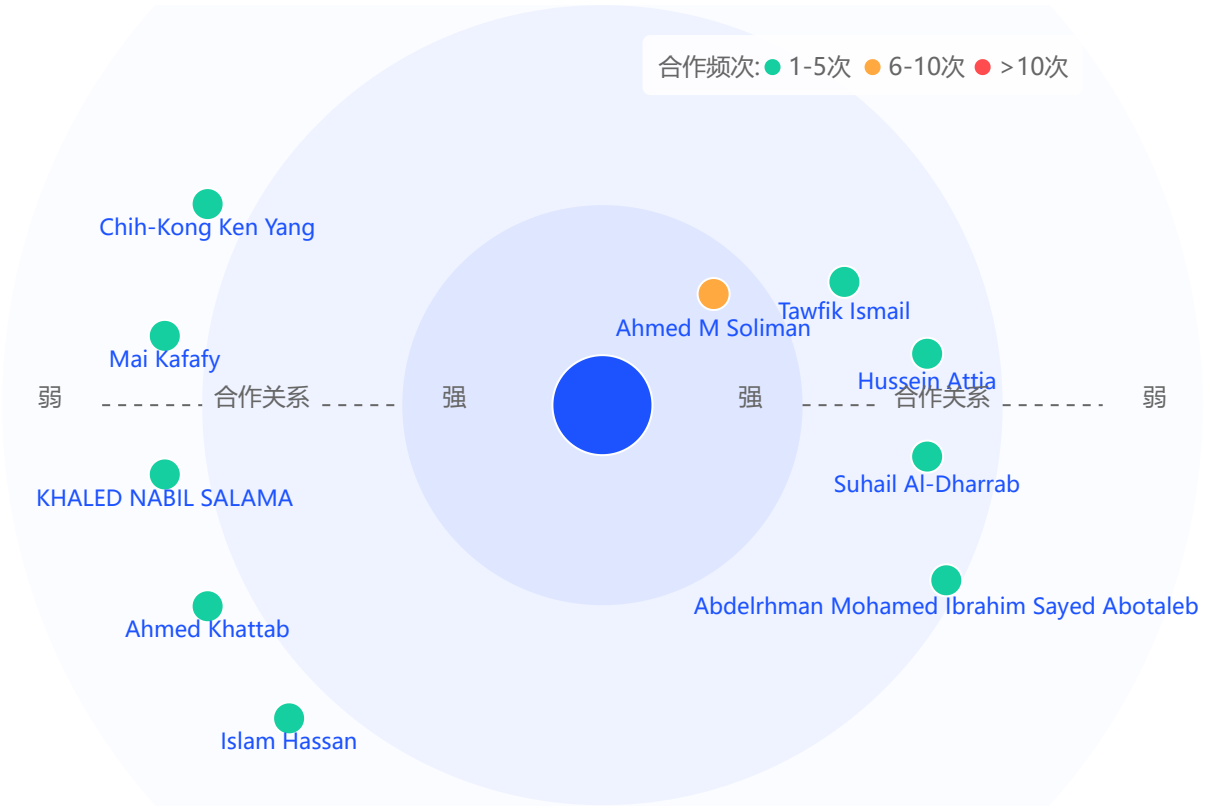
JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Mostafa, Hassan; Mohamed, Hewida; Soliman, A. M.

被引用次数: 0

2010

学术合作



Ahmed M Soliman	合作次数: 6
开罗大学	
Tawfik Ismail	合作次数: 4
开罗大学	
Hussein Attia	合作次数: 3
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